

**Extinction dynamics of fish metacommunities in
exploited dendritic networks: Guadalquivir
Mediterranean basins as a model system**

Brief summary March 2026

DONE

2

- GITHUB maintenance: Guadex

4

Mathematical formulation of the Fish Metacommunity Model This document provides a first description of the spatially-explicit modeling framework used to simulate fish metacommunity dynamics in the Guadalquivir River basin. The model is implemented as a system of Ordinary Differential Equations (ODEs) that integrate local population dynamics, environmental filtering (dams, stress, contaminants) in a 3D dendritic dispersal.

10

- General Model Structure The rate of change for the population size $N_{i,s}$ of species s at site i is defined by the sum of local dynamics and net spatial flux:

$$\frac{dN_{i,s}}{dt} = \underbrace{f_{local}(N_{i,s}, \mathbf{N}_i, E_i)}_{\text{Local Dynamics}} + \underbrace{\sum_{j \in \text{neighbors}} (m_{ji}N_{j,s} - m_{ij}N_{i,s})}_{\text{Spatial Flux}}$$

12

where:

$N_{i,s}$ is the population density of species s at site i .

14

$\mathbf{N}_i$ is the vector of all species populations at site i .

E_i represents the local environmental conditions.

16

m_{ij} is the dispersal rate from site i to site j .

18

- Local Dynamics and Environmental Filtering The local component follows a Generalized

Lotka-Volterra (GLV) structure, modified by environmental suitability filters. The local
 20 dynamics accounts for population growth and biotic interactions among species

$$f_{local} = N_{i,s} \left(r_{i,s}^{eff} - \sum_{j=1}^S \alpha_{sj} N_{i,j} \right)$$

where:

22 $r_{i,s}^{eff}$ is the effective intrinsic growth rate.

α_{sj} is the interaction coefficient representing the effect of species j on species s (asymmetric
 24 interaction matrix).

- 26 • Environmental Filtering The effective growth rate $r_{i,s}^{eff}$ is determined by the base growth rate r_s and a site-specific suitability factor $W_{i,s}$:

$$r_{i,s}^{eff} = r_s \cdot W_{i,s}$$

28 The suitability factor $W_{i,s}$ incorporates thermal tolerance and habitat quality:

$$W_{i,s} = \text{ThermalFilter}(T_i, \text{opt}_s, \sigma_s) \cdot h_i$$

- Thermal Filter We use a Gaussian-shaped "thermal window" to model species-specific
 30 temperature tolerances:

$$\text{ThermalFilter}(T_i, \text{opt}_s, \sigma_s) = \exp \left(-\frac{(T_i - \text{opt}_s)^2}{2\sigma_s^2} \right)$$

where:

32 T_i is the temperature at site i .

opt_s is the optimal temperature for species s .

34 σ_s is the thermal tolerance (standard deviation) of species s .

- 36 • Habitat Suitability h_i is the habitat suitability index at site i , which accounts for anthropogenic stressors (e.g., conductivity), river width, and mesohabitat types.
- 38 • Spatial Flux (3D Dendritic Dispersal) The spatial component models the movement of individuals through the river network, accounting for hydrological distance, elevation changes, and physical barriers. The dispersal rate m_{ij} from site i to site j is defined as:

$$m_{ij} = m \cdot \left(\frac{1}{d_{ij}} \right) \cdot x_{ij} \cdot p_{ij}$$

where:

42 m is the base dispersal intensity. * d_{ij} is the hydrological distance between site i and j . * x_{ij} is the 3D connectivity factor (elevation cost). * p_{ij} is the dam passability factor (ranging from 0.0 to 1.0).

- 46 • Elevation Cost (Upstream vs. Downstream) The cost of movement depends on the relative elevation of the sites:

* Downstream or Equal Elevation ($e_j \leq e_i$):

$$x_{ij} = 1.0$$

* Upstream ($e_j > e_i$):

$$x_{ij} = \frac{1}{1 + c(e_j - e_i)}$$

where:

48 e_i and e_j are the elevations of the source and destination sites, respectively.

c is the upstream migration cost constant.

50 • **Data Preparation** The model relies on a comprehensive data preparation pipeline, implemented in `src/datapreparation.jl` in `Guadex` to integrate heterogeneous datasets into the ODE framework. The process involves:

52 - **Species Characteristics** (load species characteristics): Loads data from `data/ABIOTIC/caracteristicas peces Guadalquivir 03-04-2018.csv`.

54 - **Thermal optima** are calculated as the midpoint of temperature ranges (**TEMPERATURE C**), and thermal breadth (σ_s) is approximated as $\sigma_s \approx \text{range}/6$. **Max size (MAX SIZE mm)** is used to scale intrinsic growth rates.

56 - **Site Data** (load dsite data): Merges connectivity data (`data/ConnectivityUTM.csv`) and environmental data (`data/ABIOTIC/Matriz Ambiental Data.csv`) on site codes (`'CODIGO'`).

58 Missing values in dam distance columns (`'Demb arr.(m)'`, `'Demb ab.(m)'`) are handled, and data is converted to numeric formats.

60 - **Interaction Matrix** (`'load interaction matrix'`): Parses qualitative interaction descriptions from `'data/BIOTIC/Interacciones peces Guadalquivir 03-04-2018 ENG.csv'` into a quantitative matrix using `'parse interaction string'`. The values are assigned based on the interaction type:

62 - "No coexist": -1.0

- "Displaces": -0.8

- "Predation": -0.5

- "Competition" or "Interfere": -0.3

- "Affects" (general): -0.2

- "Coexist" or "Neutral": 0.0

- Spatial Connectivity (build distance matrix): Constructs a sparse distance matrix from data/Matrix distances 1037puntos BRUTO FINAL.csv by grouping sites by subcatchment (CODIGO S) and sorting them by distance to the river ($Dist.Guadalq.(m)$). Connections are restricted to adjacent sites within the same subcatchment to reflect the dendritic river network.

- Environmental Filtering: Site-specific temperatures ('extract site temperatures') and habitat suitability indices ('extract habitat suitability') are extracted. Habitat suitability is normalized based on the Trophic State Index ('IET'), where higher IET values indicate lower suitability.

- Intrinsic Growth Rates ('build intrinsic growth rates'): Base growth rates are calculated as proportional to $1/\max_size$, then scaled by species presence/absence at each site (using columns ending in 'DEN' from 'data/BIOTIC/FishDensity and Juveniles Matrix.csv').

- Dispersal Matrix ('precompute dispersal matrix'):** A sparse dispersal matrix M is pre-computed, incorporating hydrological distance, elevation-dependent migration costs (using an upstream cost constant c), and dam passability factors (ranging from 0.0 to 1.0).

- Implementation Details * State Vector: The system state u is a matrix of size $N_{sites} \times N_{species}$.

Sparse Matrix Optimization: To handle large-scale river networks efficiently, the dispersal

90 rates are pre-calculated into a sparse matrix M , where M_{ji} represents the rate from i to j .

Numerical Integration: Given the potential stiffness arising from the coupling of fast
92 dispersal and slower local dynamics, the system is solved using stiff-aware solvers (e.g.,
'Rodas5' or 'TRBDF2').

94 • Model Assumptions

- Constant Interaction Matrix: Biotic interaction coefficients (α_{sj}) are assumed to be
96 constant across the entire basin, though the realized interaction depends on local densities.

- Passive/Active Dispersal Balance: Dispersal is modeled as a density-dependent flux
98 proportional to the population at the source site.

- Gaussian Thermal Niche:** Species performance is assumed to decay symmetrically as
100 temperatures deviate from the optimum.

- Dam Passability: Dams act as semi-permeable filters that reduce the effective dispersal
102 rate between connected nodes.

1 TODO

104 • RUN simulations basin under global change scenario, calibration and uncertainty estimations

• MODEL implementation using last version of all the datasets

106 • RUN simulations basin and subbasin global change scenario using last version of all the
datasets

108 • GIT Functional repository to reproduce all simulations

• ANALYSIS outputs global change scenario

- 110 • KNOWLEDGE transfer interface
- WRITING reporting
- 112 • SUBMISSION paper to journal